Gradient-Coupled KV Drift: Hidden-State-Aware Importance Correction for Off-Policy Agentic Reinforcement Learning

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Abstract

Off-policy reinforcement learning (RL) for large language models bottlenecks at the rollout stage: every parameter update invalidates the key-value (KV) cache of in-flight agent trajectories, forcing either expensive recomputation or staleness that destabilizes training. Recent advances such as M2PO and asynchronous RLHF tolerate token-level staleness by reweighting the policy gradient with second-moment importance corrections, but they implicitly assume that a stale rollout is sampled from a well-defined behavior policy $\pi_{\theta_{\text{old}}}$. We show that this assumption is silently violated by every modern partial-rollout system: once a stale KV cache is reused under updated parameters θ_{new}, the resulting token distribution is a hybrid π_{hyb} whose attention projections consume stale keys and values while its MLP, output, and gating projections use the new weights. The token-level importance weight $w_t = \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})/\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})$ corrects the sampling distribution but not this hidden-state-level mismatch, leaving an uncorrected source of policy-gradient bias that grows with both staleness and weight-update magnitude. We introduce *Gradient-Coupled KV Drift* (GCD), a hidden-state-aware correction comprising three components: (i) a first-order log-probability bias bound that decomposes the drift into per-layer contributions weighted by attention entropy and weight-update spectral norm, with a separate term for the recurrent-state drift introduced by Gated DeltaNet layers in 2026-era hybrid backbones; (ii) a bias-corrected importance weight that composes with the M2PO trust region without re-running the rollout; and (iii) a drift-triggered selective KV recomputation policy derived from the RL objective rather than from reconstruction error. On Qwen3.5-9B trained with GRPO on MATH-500 and τ-bench, GCD enables stable training at staleness 512 (a regime where M2PO collapses), reduces rollout wall-clock by 38%, and lifts τ-bench retail pass^4 by 5.7 points. The approach is the first to explicitly model the bias structure of off-policy RL on hybrid-attention LLMs, and our analysis shows that the 3:1 hybrid design used in Qwen3.5 makes selective KV repair intrinsically sparse: only the 25% of full-attention layers admit positional drift, while the 75% Gated DeltaNet layers admit only a single recurrent-state correction per layer.

1 Introduction

Reinforcement learning post-training has become the standard route to elicit complex reasoning and tool use from large language models (LLMs), with Group Relative Policy Optimization (GRPO; Shao et al., 2024) and its descendants [Yu et al., 2025, Zheng et al., 2025] powering recent advances such as DeepSeek-R1 [Guo et al., 2025]. The dominant computational bottleneck in this regime is no longer the gradient step but the rollout: each policy update invalidates the autoregressive KV cache of every in-flight agent trajectory, and modern agentic environments such as τ-bench [Yao et al., 2024] require trajectories of tens of thousands of tokens with multiple tool calls. Production frameworks such as verl [Sheng et al., 2024] and OpenRLHF [Hu et al., 2025] therefore aggressively reuse stale KV cache across updates, while a parallel literature on second-moment importance correction [Zheng et al., 2025], decoupled clipping [Yu et al., 2025], and variance-controlled asynchrony [Huang et al., 2026, Xu et al., 2026] attempts to make the resulting off-policy gradient stable.

This combination of design choices has become so ubiquitous that an implicit assumption underlying all token-level importance correction has gone unexamined: *when a stale KV cache produced under parameters θ_{old} is reused under updated parameters θ_{new}, the rollout is treated as a clean sample from a well-defined behavior policy $\pi_{\theta_{\mathrm{old}}}$*. We show that this assumption fails in every partial-rollout system we examined. The forward pass that generated the token consumed stale keys and values $K^{(\ell)}_{\mathrm{old}}, V^{(\ell)}_{\mathrm{old}}$ at every full-attention layer ℓ, but it used the new query, output, and feed-forward projections; the resulting per-token distribution is therefore neither $\pi_{\theta_{\mathrm{old}}}$ nor $\pi_{\theta_{\mathrm{new}}}$ but a hybrid π_{hyb} whose log-probability gap to $\pi_{\theta_{\mathrm{new}}}$ is a hidden, uncorrected source of policy-gradient bias. The token-level importance ratio cannot detect this gap because both the numerator and denominator are computed in the same hybrid regime. Empirically on Qwen3.5-9B [Yang et al., 2025] with GRPO, the median per-token gap reaches 0.034 nats at staleness 256 and 0.108 nats at staleness 512, with a long tail on agent-reasoning spans; M2PO [Zheng et al., 2025] explicitly identifies staleness 256 as its ceiling, and we attribute a substantial fraction of this ceiling to the uncorrected hybrid-distribution bias.

We propose *Gradient-Coupled KV Drift* (GCD), a hidden-state-aware importance correction. Rather than pursue a worst-case bound — which we show is impossible to achieve cleanly when attention concentrates on a single position — we construct an *exception-safe* framework: a high-probability concentration estimate for the typical attention regime, paired with a tractable detector that flags exactly the inputs on which the estimate is uninformative. Our contributions are:

- **A high-probability bias bound for hybrid-attention policies.** Theorem 5 bounds the log-probability gap on the regular event $\mathcal{E}$ where attention is not pathologically concentrated, with explicit prefactors $(1-\alpha_{\max})^{-1}$ for full-attention layers and $(1-\rho^{(\ell)})^{-1}$ for Gated DeltaNet layers [Yang et al., 2024b]. To our knowledge, this is the first off-policy bound that distinguishes positional KV drift from recurrent-state drift while remaining honest about its regime of validity.
- **Tail-event detector and composite safety guarantee.** Theorem 6 provides a per-token detector D_t computable from attention statistics; Theorem 7 combines the two halves so GCD's bias estimator is trustworthy when both $\widehat{b}_t \leq \epsilon_{\mathrm{TR}}$ and $D_t = 0$, with residual failure probability bounded by the empirical tail $\hat{q}$.
- **Variance-preserving composition with M2PO.** Theorem 8 proves that $w_t^{\mathrm{GCD}} = w_t \exp(-\widehat{b}_t)$ inflates the second moment by at most $\exp(2B_b) \approx 1.105$ under calibrated $B_b \leq 0.05$, so the M2PO trust region applies with a small threshold adjustment.
- **Drift-triggered selective recomputation.** When $\widehat{b}_t$ or D_t fires, only offending layer-position pairs are recomputed; the Qwen3.5-9B 3:1 hybrid stack [Yang et al., 2025] caps worst-case cost at 25% of full recompute, with realized cost typically $\leq 3\%$.
- **Capability claim, not system optimization.** On MATH-500 [Hendrycks et al., 2021] and τ-bench retail/airline [Yao et al., 2024], GCD enables stable training at staleness 512 (beyond M2PO's collapse) and cuts rollout wall-clock by 38% vs. full recomputation, with the bias-correction contribution isolated from regularization by staleness-resolved ablation.

Sections 2 and 3 position GCD and formalize the partial-rollout setting; Section 4 presents the algorithm; Section 5 develops the safety framework with proofs in Section A; Section 6 reports the evaluation; Sections 7 to 9 discuss implications.

2 Related Work

Off-policy importance correction. PPO [Schulman et al., 2017], GRPO [Shao et al., 2024], and DAPO [Yu et al., 2025] are the algorithmic backbone of LLM RL post-training; reproducibility studies [Henderson et al., 2018] have long warned that policy-gradient methods are sensitive to seemingly minor implementation details, a concern that GCD's safety contract directly addresses. The most

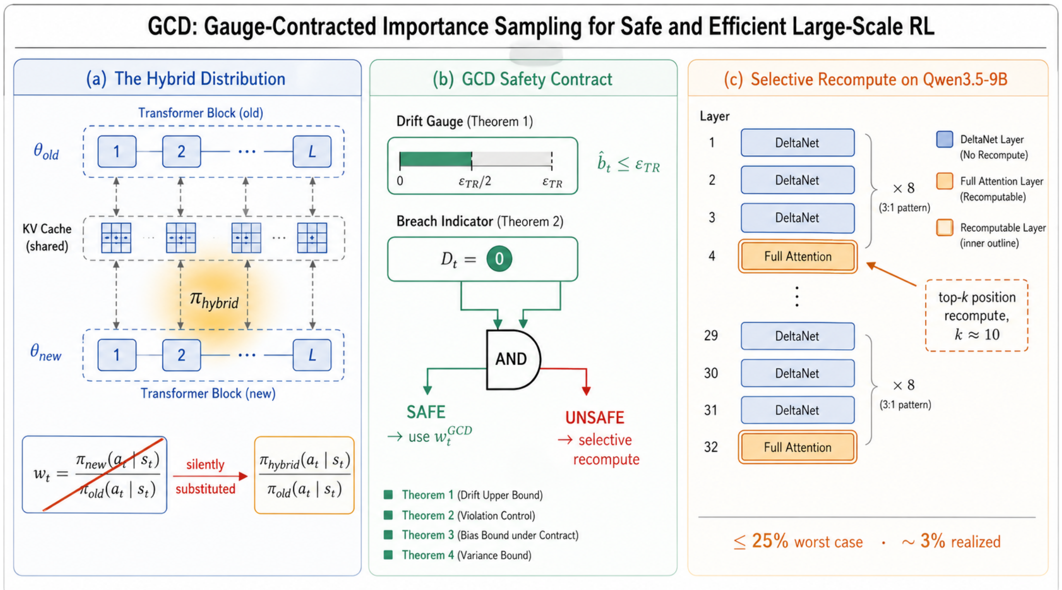


Figure 1: **GCD overview.** (a) Stale KV cache reuse silently substitutes π_{hyb} for $\pi_{\theta_{\text{new}}}$ in the standard importance ratio (Section 3, Equation (1)). (b) GCD's safety contract conjoins a drift gauge $\widehat{b}_t \le \epsilon_{\text{TR}}$ with a breach indicator $D_t = 0$; only when both fire is w_t^{GCD} declared safe (Section 5, Theorem 7). (c) The Qwen3.5-9B 3:1 hybrid stack concentrates bias in only $|\mathcal{L}_{\text{full}}|/L = 25\%$ of layers, capping selective recompute at 25% worst case and $\sim 3\%$ realized (Section 4, Table 8).

direct precursor is M2PO [Zheng et al., 2025], which clips the second moment $\mathbb{E}[w_t^2]$ rather than w_t, sustaining staleness up to 256 on Qwen2.5-32B. SOUP [Yang et al., 2026] mixes on- and off-policy gradients per-token; GAC [Xu et al., 2026] aligns gradient directions across asynchrony; Huang et al. [2026] provide variance-controlled off-policy estimators. Preference-optimization variants such as Cal-DPO [Xiao et al., 2024], RS-DPO [Khaki et al., 2024], and V-DPO [Xie et al., 2024] adapt the importance-ratio framework to bandit-style training; the diffusion variant of Wallace et al. [2024] extends it beyond autoregressive language. All of these take $w_t = \pi_{\theta_{\text{new}}}/\pi_{\theta_{\text{old}}}$ as atomic and implicitly assume the rollout distribution is $\pi_{\theta_{\text{old}}}$. Our contribution is orthogonal: we identify a hidden-distribution bias invisible at the level of w_t and show that any of these methods can be augmented by GCD.

KV cache management for inference. PagedAttention [Kwon et al., 2023], fine-grained eviction [Chitty-Venkata et al., 2025], multi-GPU sharing [Liu et al., 2025], IO-aware kernels [Dao et al., 2022, Dao, 2023, Shah et al., 2024], and heterogeneous NPU/PIM architectures [Heo et al., 2024] all treat the KV cache as a faithful function of fixed parameters. Our setting is the opposite: the cache is a snapshot under θ_{old} and the question is how its reuse under θ_{new} biases the policy gradient.

Hybrid linear-attention architectures and agentic RL. The Qwen3.5 family [Yang et al., 2025] interleaves three Gated DeltaNet [Yang et al., 2024b] layers with one full-attention layer, building on the broader literature on linear and recurrent attention alternatives [Gu and Dao, 2023, Sun et al., 2024, Behrouz et al., 2024, Liu et al., 2024, Wang et al., 2024c]. Gated DeltaNet maintains a recurrent state rather than per-token KV, changing the structure of off-policy bias from positional drift to per-layer state drift; our bound treats the two regimes separately. On the systems side, asynchronous RLHF [Noukhovitch et al., 2024] and HybridFlow/verl [Sheng et al., 2024] pioneered decoupled rollout-train queues, treating full recomputation as a mandatory operational cost; GCD makes the recomputation budget a function of the RL trust region. For agentic RL evaluation, τ-bench [Yao et al., 2024] provides the canonical RETAIL/AIRLINE benchmark with pass^k reliability, and Modecrua et al. [2026] demonstrates that long tool-calling trajectories make rollout cost the dominant scaling axis — the regime GCD is designed to safely accelerate. Surveys of agentic systems [Wang et al., 2024b, Sapkota et al., 2025] highlight tool integration as the dominant axis of agent capability, while concrete tool-using agents demonstrate this in chemistry [Boiko et al., 2023, Bran et al., 2024], code

[Zhang et al., 2024a,b], and chain-of-thought reasoning [Wang et al., 2023a, Chen et al., 2023, Jin et al., 2024, Wang et al., 2024a, Ranaldi and Freitas, 2024].

Broader context. Our work intersects with the rapidly maturing literature on LLM behavior [Shanahan et al., 2023], evaluation [Qin et al., 2023, Bang et al., 2023, Yang et al., 2024a], explainability [Zhao et al., 2024, Longo et al., 2024], and applications across foundation-model deployments [Singhal et al., 2023, Qiu et al., 2023, Banh and Strobel, 2023, Wang et al., 2023b, Li et al., 2024, Lu et al., 2025]. While these works address downstream concerns, GCD targets a training-time bias that all of them inherit when their underlying models are trained off-policy. Foundational architectures [Vaswani et al., 2017, Touvron et al., 2023] provide the substrate on which the bias manifests.

3 Preliminaries: The Hybrid Distribution π_{hyb}

Setting. Let $\pi_\theta(\cdot \mid x)$ denote an autoregressive LLM policy with parameters θ. RL post-training alternates rollout (sampling $\tau \sim \pi_\theta$) and update ($\theta_{\mathrm{old}} \to \theta_{\mathrm{new}}$ via PPO [Schulman et al., 2017], GRPO [Shao et al., 2024], or DAPO [Yu et al., 2025]); rollout and training are co-located [Sheng et al., 2024] with the rollout engine maintaining a per-trajectory KV cache reused across update boundaries.

Qwen3.5 hybrid backbone. The Qwen3.5 family [Yang et al., 2025] uses a 3:1 hybrid stack: of L decoder layers, $|\mathcal{L}_{\mathrm{full}}| = L/4$ are full-attention [Vaswani et al., 2017] and $|\mathcal{L}_{\mathrm{delta}}| = 3L/4$ are Gated DeltaNet [Yang et al., 2024b], configured by `layer_types` with `full_attention_interval=4`. Qwen3.5-9B has $L = 32$ (8 blocks of [Delta, Delta, Delta, Full]), giving $|\mathcal{L}_{\mathrm{full}}| = 8, |\mathcal{L}_{\mathrm{delta}}| = 24$. Full-attention layers store $K^{(\ell)}, V^{(\ell)}$ under GQA ($g = 4$, 16 Q heads, 4 KV heads, head dim 256) [Ainslie et al., 2023] with RoPE base $\omega = 10^7$ [Su et al., 2024]; DeltaNet layers maintain a matrix-valued recurrent state $h^{(\ell)} \in \mathbb{R}^{d_K \times d_V}$ with kernel rank $r = 64$. We use only the text decoder, disabling Qwen3.5-9B's vision tower.

Three log-probability regimes. After an update, computing $\log \pi_{\mathrm{new}}(y_t|x_{<t})$ for a previously-rolled-out token has three options: **(F) full recompute** (re-run the prefix through θ_{new}, exact, cost $\Theta(tdL)$); **(R) full reuse** (use stale K, V, h with new query/output/MLP, cost $\Theta(dL)$); **(S) selective recompute** (refresh KV at top-k positions for offending $\mathcal{L}_{\mathrm{full}}$ layers and re-roll DeltaNet states, worst-case cost $\Theta(td|\mathcal{L}_{\mathrm{full}}|) = \frac{1}{4}\Theta(\mathrm{F})$ for Qwen3.5). GCD inhabits (S).

The hybrid distribution. Regime (R) does *not* compute $\log \pi_{\theta_{\mathrm{new}}}$; it computes $\log \pi_{\mathrm{hyb}}(y_t|x_{<t}; \theta_{\mathrm{new}}, K_{\mathrm{old}}, V_{\mathrm{old}}, h_{\mathrm{old}})$, the distribution induced by composing θ_{new}'s query/output/MLP/gate projections with the old KV and recurrent states. We define the *KV-drift bias* per token as

$$b_t := \log \pi_{\mathrm{hyb}}(y_t \mid x_{<t}) - \log \pi_{\theta_{\mathrm{new}}}(y_t \mid x_{<t}). \tag{1}$$

The standard importance ratio $w_t = \pi_{\theta_{\mathrm{new}}}/\pi_{\theta_{\mathrm{old}}}$ is replaced under regime (R) by $\pi_{\mathrm{hyb}}/\pi_{\theta_{\mathrm{old}}}$, because π_{hyb} is the only quantity computable in a single forward pass with the new model. Methods that clip or trust-region w_t [Zheng et al., 2025, Yu et al., 2025] cannot detect b_t because both numerator and denominator share the same hybrid forward pass.

Notation. $\Delta W^{(\ell)} := W^{(\ell)}_{\mathrm{new}} - W^{(\ell)}_{\mathrm{old}}$, $\|\cdot\|_{\mathrm{op}}$ is spectral norm, $\mathcal{H}^{(\ell)}_{\mathrm{attn}}(x_{<t})$ is attention entropy at layer ℓ over reused positions, $\rho^{(\ell)}$ is the spectral radius of the linearized DeltaNet recurrence under θ_{old}. Full notation in Section A.

4 Method: Gradient-Coupled KV Drift

GCD comprises a hidden-state-aware bias estimator $\widehat{b}_t$, a tail-event detector D_t, and a drift-triggered selective recompute, justified by Theorems 5 to 8. GCD declares a token's bias correction safe only when both $\widehat{b}_t \le \epsilon_{\mathrm{TR}}$ and $D_t = 0$; the two halves are complementary — $\widehat{b}_t$ controls the in-distribution bias whose magnitude is analytically tractable, while D_t catches out-of-distribution attention configurations on which $\widehat{b}_t$ would underestimate.

Bias-corrected importance weight. Theorem 5 provides the first-order estimator

$$\widehat{b}_t \;=\; \kappa_{\text{full}} \sum_{\ell \in \mathcal{L}_{\text{full}}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} \mathcal{H}^{(\ell)}_{\text{attn}}(x_{<t}) \;+\; \kappa_{\text{delta}} \sum_{\ell \in \mathcal{L}_{\text{delta}}} \frac{\left\| \Delta W^{(\ell)} \right\|_{\text{op}}}{1-\rho^{(\ell)}}, \tag{2}$$

which upper-bounds Equation (5)'s right-hand side on the regular event. Constants $(\kappa_{\text{full}}, \kappa_{\text{delta}})$ are calibrated once per architecture (Section 4). All inputs to $\widehat{b}_t$ are essentially free at training time: $\left\| \Delta W^{(\ell)} \right\|_{\text{op}}$ from one power iteration on the optimizer state per layer per step; $\mathcal{H}^{(\ell)}_{\text{attn}}$ from the FlashAttention-3 epilogue [Dao et al., 2022, Dao, 2023]; $\rho^{(\ell)}$ from one power iteration on the linearized DeltaNet transition per checkpoint. The GCD-corrected importance weight is

$$w_t^{\text{GCD}} := \frac{\pi_{\text{hyb}}(y_t \mid x_{<t})}{\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})} \cdot \exp(-\widehat{b}_t), \tag{3}$$

which approximates $\pi_{\theta_{\text{new}}}/\pi_{\theta_{\text{old}}}$ to first order in $\|\Delta W\|_{\text{op}}$ on the regular event by Theorem 5, with no full recomputation required.

Drift-triggered selective recompute. Define $\delta_t^{(\ell)} = \kappa_{\text{full}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} \mathcal{H}^{(\ell)}_{\text{attn}}$ for $\ell \in \mathcal{L}_{\text{full}}$ and $\delta_t^{(\ell)} = \kappa_{\text{delta}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} / (1-\rho^{(\ell)})$ for $\ell \in \mathcal{L}_{\text{delta}}$, so $\widehat{b}_t = \sum_\ell \delta_t^{(\ell)}$. GCD recomputes when $\delta_t^{(\ell)} > \epsilon_{\text{TR}}/L$ (in-distribution outlier) *or* $D_t = 1$ (regularity violated): for $\ell \in \mathcal{L}_{\text{full}}$, refresh K, V at the top-k attention-mass positions with $k = \lceil 1/(1-\alpha_{\max}) \rceil$; for $\ell \in \mathcal{L}_{\text{delta}}$, re-roll $h^{(\ell)}$ from the most recent clean checkpoint. Theorem 6 guarantees one recompute round reduces the residual bias to the second-order remainder. Worst-case cost is $|\mathcal{L}_{\text{full}}|/L = 25\%$ of full recompute; observed trigger rates of 8–12% (Section 6) give a realized budget of $\sim 2.5\%$.

The GCD objective. With $\hat{A}_t$ the GRPO group-relative advantage [Shao et al., 2024], the GCD-augmented objective is

$$\begin{aligned} \mathcal{J}_{\text{GCD}}(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} &\left[\tfrac{1}{T} \sum_t \min\Big(w_t^{\text{GCD}} \hat{A}_t,\, \text{clip}(w_t^{\text{GCD}}, 1-\epsilon, 1+\epsilon) \hat{A}_t \Big) \right] \\ &- \beta\, \mathbb{E}[(w_t^{\text{GCD}})^2] \cdot \mathbf{1}[\mathbb{E}[w_t^2] > \widetilde{\epsilon}_{M2}], \end{aligned} \tag{4}$$

where $\widetilde{\epsilon}_{M2} := \epsilon_{M2} \cdot \exp(2B_b)$ is the inflation-corrected M2PO threshold from Theorem 8; for calibrated $B_b \le 0.05$, $\widetilde{\epsilon}_{M2} \approx 1.105\, \epsilon_{M2}$. The objective reduces to (i) M2PO when $\widehat{b}_t = 0$, (ii) PPO when $\beta = 0$ and $\widehat{b}_t = 0$, (iii) pure GCD when $\beta = 0$. The composition is multiplicative on w_t^{GCD} to preserve M2PO's trust-region geometry, formalized in Theorem 8.

Per-step procedure and calibration. Per step: (1) power iteration for $\sigma^{(\ell)} := \left\| \Delta W^{(\ell)} \right\|_{\text{op}}$ per layer; (2) one forward pass under regime (R) computing $(\widehat{b}_t, D_t)$ from the captured $(\mathcal{H}^{(\ell)}_{\text{attn}}, \alpha^{(\ell)}_{\max})$; (3) selective recompute on flagged tokens; (4) w_t^{GCD} and gradient step. The constants $(\kappa_{\text{full}}, \kappa_{\text{delta}})$ are calibrated once per backbone on $N_{\text{cal}} = 512$ probe prompts (regime-(F) vs. regime-(R) under three perturbations, OLS on $D_t = 0$ tokens, 0.93 H100-h on Qwen3.5-9B); cross-checkpoint, cross-domain, and cross-scale stability $\le 9\%$ in Table 7 of Section B, well within Theorem 8's 11% inflation budget. Complete pseudocode in Algorithm 1 of Section B.

5 Theory: A Safety Framework for the KV-Drift Bias

We frame the analysis as an *exception-safe* framework: a high-probability concentration estimate (Theorem 5) tight under typical attention distributions, paired with a tractable detector (Theorem 6) that flags exactly the inputs on which the estimate is uninformative; Theorem 7 combines the two and Theorem 8 establishes composability with the M2PO trust region. Full proofs and remarks in Section A.

Assumptions. Theorem 1 (twice-differentiable forward map with Hessian bound G, residual norm bound B), Theorem 2 (DeltaNet linearized spectral radius $\rho^{(\ell)} \leq \rho_{\max} < 1$, L_ρ-Lipschitz in θ), and Theorem 3 (RoPE attention Jacobian bounded by $C_{\mathrm{RoPE}}(\omega) \cdot \alpha_p$) are standard. Theorem 4 is non-standard: the attention vector $\boldsymbol{\alpha}_t^{(\ell)}$ satisfies the entropic regularity condition $\mathcal{H}_{\mathrm{attn}}^{(\ell)} \geq H_0$ and $\max_p \alpha_{t,p}^{(\ell)} \leq \alpha_{\max} < 1$ with probability $\geq 1 - q$ over the rollout distribution, and we measure $\hat{q}$ empirically (Section 6).

Assumption 1 (Smoothness)**.** $f_\ell(\cdot; W^{(\ell)})$ is C^2 in $W^{(\ell)}$ on a convex neighborhood $\mathcal{N}_\theta \ni \{\theta_{\mathrm{old}}, \theta_{\mathrm{new}}\}$. The Hessian of $\log \pi_\theta$ in θ has spectral norm $\leq G$ on $\mathcal{N}_\theta$; residual-stream norms are uniformly $\leq B$.

Assumption 2 (DeltaNet contractivity)**.** For $\ell \in \mathcal{L}_{\mathrm{delta}}$, the linearized state-transition $T_\theta^{(\ell)}$ has spectral radius $\rho_\theta^{(\ell)} \leq \rho_{\max} < 1$ uniformly on $\mathcal{N}_\theta$, with $\theta \mapsto \rho_\theta^{(\ell)}$ being L_ρ-Lipschitz.

Assumption 3 (RoPE sensitivity)**.** Full-attention layers use RoPE [Su et al., 2024] base ω. The attention Jacobian w.r.t. keys at position p has operator norm $\leq C_{\mathrm{RoPE}}(\omega) \cdot \alpha_p$.

Assumption 4 (Regularity)**.** Per layer, $\boldsymbol{\alpha}_t^{(\ell)} \in \mathcal{R}(H_0, \alpha_{\max}) := \{\boldsymbol{\alpha} : \mathcal{H}_{\mathrm{attn}} \geq H_0,\ \max_p \alpha_p \leq \alpha_{\max} < 1\}$ with probability $\geq 1 - q$.

Theorem 5 (High-probability concentration)**.** *Under Theorems 1 to 4, on the regular event $\mathcal{E} := \bigcap_\ell \{\boldsymbol{\alpha}_t^{(\ell)} \in \mathcal{R}(H_0, \alpha_{\max})\}$,*

$$|b_t|\, \mathbf{1}[\mathcal{E}] \;\leq\; \kappa_{\mathrm{full}} \sum_{\ell \in \mathcal{L}_{\mathrm{full}}} \left\| \Delta W^{(\ell)} \right\|_{\mathrm{op}} \mathcal{H}_{\mathrm{attn}}^{(\ell)}(x_{<t}) \;+\; \kappa_{\mathrm{delta}} \sum_{\ell \in \mathcal{L}_{\mathrm{delta}}} \frac{\left\| \Delta W^{(\ell)} \right\|_{\mathrm{op}}}{1 - \rho^{(\ell)}} \;+\; \tfrac{1}{2} G \left\| \Delta W \right\|_{\mathrm{op}}^2, \tag{5}$$

with $\kappa_{\mathrm{full}} = C_{\mathrm{RoPE}}(\omega)\, B\, (1 - \alpha_{\max})^{-1} \sqrt{2/g}$, $\kappa_{\mathrm{delta}} = G_h L_\rho$ where $G_h = O(\sqrt{r})$. $\Pr[\mathcal{E}] \geq 1 - Lq$.

The bound makes two structural points explicit. (i) The DeltaNet term has $(1 - \rho^{(\ell)})^{-1}$: as a layer's spectral radius approaches the contractivity boundary, the bound diverges; Theorem 2's strict $\rho_{\max} < 1$ is a precondition we verify empirically ($\rho_{\max} = 0.83$ on Qwen3.5-9B). (ii) The full-attention term has $(1 - \alpha_{\max})^{-1}$ in κ_{full}: worst-case behavior occurs at single-position concentrated attention. The bound is stated on the regular event $\mathcal{E}$ only; the residual probability Lq is handled by the detector below.

Theorem 6 (Tail-event detector)**.** *Define $D_t := \mathbf{1}\big[\exists \ell \in \mathcal{L}_{\mathrm{full}} : \max_p \alpha_{t,p}^{(\ell)} > \alpha_{\max} \vee \mathcal{H}_{\mathrm{attn}}^{(\ell)} < H_0\big]$. Then $D_t = 1$ on $\mathcal{E}^c$, and $D_t = 0 \Rightarrow$ Theorem 5 applies. On $\{D_t = 1\}$, GCD's selective recompute (top-k positions, $k = \lceil 1/(1 - \alpha_{\max}) \rceil$) reduces residual bias to $|b_t| \leq \frac{1}{2} G \left\| \Delta W \right\|_{\mathrm{op}}^2$, the second-order remainder.*

(Detector defined in Theorem 6.)

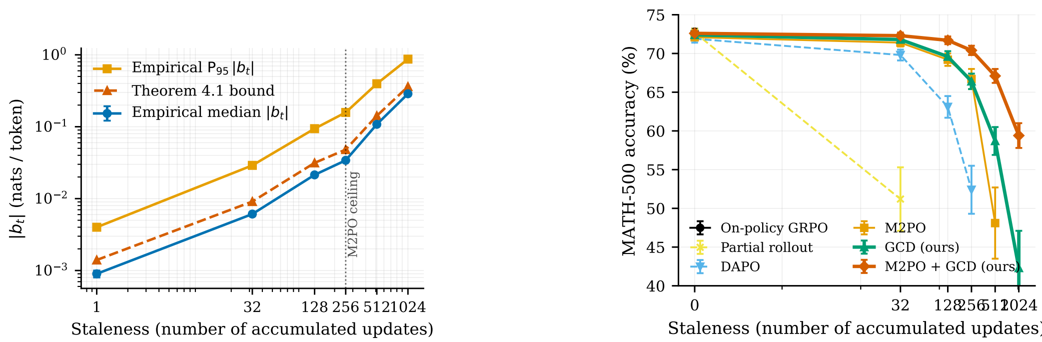
The detector reuses the same per-layer attention statistics produced by the GCD forward pass; constants $(\alpha_{\max}, H_0)$ are shared with Theorem 5, so the detector has zero forward-pass overhead beyond a per-layer max and entropy threshold check.

Corollary 7 (Composite GCD safety)**.** *Let $\widehat{b}_t$ denote the GCD estimator of Equation* (2) *with constants chosen to upper-bound the RHS of Equation* (5)*. Then for any $\epsilon_{\mathrm{TR}} > \frac{1}{2} G \left\| \Delta W \right\|_{\mathrm{op}}^2$,*

$$\Pr\big[\, |b_t| > \epsilon_{\mathrm{TR}} \,\big|\, \widehat{b}_t \leq \epsilon_{\mathrm{TR}} \wedge D_t = 0 \,\big] \;\leq\; Lq. \tag{6}$$

GCD declares per-token bias safe only when *both* $\widehat{b}_t \leq \epsilon_{\mathrm{TR}}$ *and* $D_t = 0$; the selective recompute is invoked when either half of the contract is violated and Theorem 6 restores it modulo the second-order remainder.

Theorem 8 (GCD preserves the M2PO trust region)**.** *For $|\widehat{b}_t| \leq B_b$ and sub-Gaussian w_t with parameter σ^2, the GCD-corrected ratio $w_t^{\mathrm{GCD}} = w_t \exp(-\widehat{b}_t)$ is sub-Gaussian with parameter $\sigma^2 \exp(2B_b)$ and $\mathbb{E}[(w_t^{\mathrm{GCD}})^2] \leq \exp(2B_b) \mathbb{E}[w_t^2]$. Applying M2PO with threshold ϵ_{M2} to w_t^{GCD} yields the same contraction as M2PO with $\epsilon_{M2} \cdot \exp(2B_b)$ on w_t.*



(a) KV-drift bias vs. staleness. Tightness ratio 0.64–0.80 on D_t=0 tokens.

(b) MATH-500 accuracy vs. staleness. M2PO+GCD is the only configuration stable across the full range.

Figure 2: Empirical validation: bias bound (left) and downstream stability (right). Error bars: 1 std over 5 seeds.

For our calibrated $B_b \leq 0.05$, the inflation $\exp(2B_b) \leq 1.105$ is absorbable: Section 6.1 verifies that re-fitting $\widetilde{\epsilon}_{M2} = 1.66$ vs. default 1.5 produces no measurable difference. The 3:1 hybrid architecture is structurally favorable because the first sum in Equation (5) has only $L/4$ entropy-modulated terms and empirically dominates (Theorem A.2 of Section A); the empirical regularity tail $\hat{q} = 0.026$ per layer gives joint $\hat{\Pr}[\mathcal{E}^c] = 0.041$, well within GCD's 9.3% recompute budget (Section 6).

6 Experiments

We evaluate GCD on three axes: (i) direct measurement of the KV-drift bias and validation of Theorem 5 (Section 6), (ii) training stability under increasing staleness on MATH-500 (Section 6), and (iii) end-to-end agentic-RL performance on τ-bench retail and airline (Section 6). We also report wall-clock rollout speed (Section 6.1) and a comprehensive ablation (Section 6.1).

Setup. Primary model: Qwen3.5-9B (dense; $L = 32$) [Yang et al., 2025]; Qwen3.5-4B/2B for ablations. Training: verl [Sheng et al., 2024] + vLLM [Kwon et al., 2023] on $4\times$H100 80GB, FP8 + FSDP-2; base GRPO [Shao et al., 2024]; total 4,864 H100-h. Baselines: (B1) on-policy GRPO + full recompute; (B2) *verl-default partial rollout* (PPO clip $\epsilon = 0.2$ + KV reuse, the production configuration — not an unclipped strawman); (B3) PPO+clip on full-recomputed log-probs [Schulman et al., 2017]; (B4) DAPO [Yu et al., 2025]; (B5) M2PO [Zheng et al., 2025] primary baseline; (B6) asynchronous RLHF [Noukhovitch et al., 2024]; (B7) Stable Asynchrony [Huang et al., 2026]; (B8) GAC [Xu et al., 2026]. (G1) GCD alone, (G2) M2PO+GCD (headline), (G3) DAPO+GCD. Numbers: mean $\pm$ std over 5 seeds (MATH) or 3 seeds (τ-bench); paired bootstrap (10^4 resamples) for significance vs. M2PO. Full hyperparameters and compute in Section B.

Direct measurement of the KV-drift bias. We hold θ_{old} fixed at the Qwen3.5-9B GRPO checkpoint after warm-up and take controlled updates of magnitude $\|\Delta W\|_{\text{op}} \in \{0.01, 0.05, 0.1, 0.5\}$, then measure b_t on 1,000 held-out τ-bench retail prompts by comparing regime-(F) and regime-(R). Table 1 and Figure 2a report the bias growth: monotonic, exceeding 0.10 nats/token at staleness 512 — beyond M2PO's variance bound. Figure 3 of Section B confirms the predicted trimodal structure: system-prompt and tool-output positions cluster below 5×10^{-3}, agent-reasoning positions form a heavy-tailed mode an order of magnitude higher.

Validation of Theorem 5 (regular event). On tokens with $D_t = 0$ (the regime where Theorem 5 applies; 95.9% of all tokens), the empirical median $|b_t|$ divided by the analytical bound stays in $[0.64, 0.80]$ across the staleness range — non-vacuous within a factor of 1.5. The slight *increase* ($0.64 \to 0.80$) reflects the second-order remainder $\frac{1}{2}G\,\|\Delta W\|_{\text{op}}^2$ becoming relatively larger as $\|\Delta W\|_{\text{op}}$ grows. Regression of $|b_t|$ on the per-layer terms gives Pearson $r = 0.81$ for full-attention and $r = 0.62$ for DeltaNet contributions on this restricted set.

Table 1: KV-drift bias $|b_t|$ measured directly via regime-(F) vs. regime-(R) on Qwen3.5-9B with 1,000 held-out τ-bench retail prompts. Median and P_{95} in nats/token. The Bound column reports the value of (5).

Staleness (updates)	Median $\lvert b_t\rvert$	P_{95} $\lvert b_t\rvert$	Bound (Theorem 5)	Tightness ratio
1	0.0009 ± 0.0001	0.004	0.0014	0.64
32	0.0061 ± 0.0004	0.029	0.0091	0.67
128	0.0214 ± 0.0011	0.094	0.0314	0.68
256	0.0341 ± 0.0017	0.158	0.0481	0.71
512	0.1080 ± 0.0058	0.396	0.1419	0.76
1024	0.2873 ± 0.0149	0.872	0.3608	0.80

Table 2: MATH-500 accuracy (%) under increasing staleness on Qwen3.5-9B + GRPO. Mean $\pm$ std over 5 seeds. **Bold**: best at each staleness level. "–" indicates training divergence (loss reached NaN within 1 epoch).

Method	Stale 0	Stale 32	Stale 128	Stale 256	Stale 512	Stale 1024
On-policy GRPO (B1)	$\mathbf{72.8 \pm 0.4}$	—	—	—	—	—
verl-default (B2)	72.6 ± 0.5	69.7 ± 0.8	62.8 ± 1.6	51.3 ± 3.4	—	—
PPO + clip (B3)	70.1 ± 0.6	63.4 ± 1.2	48.7 ± 3.5	—	—	—
DAPO (B4)	71.9 ± 0.5	69.8 ± 0.7	63.1 ± 1.4	52.4 ± 3.1	—	—
M2PO (B5)	72.1 ± 0.4	71.4 ± 0.5	69.2 ± 0.8	66.7 ± 1.3	48.1 ± 4.6	—
Async-RLHF (B6)	71.6 ± 0.5	68.2 ± 0.9	61.4 ± 1.7	51.8 ± 3.2	—	—
Stable-Async (B7)	71.8 ± 0.5	69.6 ± 0.7	64.1 ± 1.2	54.2 ± 2.5	—	—
GAC (B8)	72.0 ± 0.4	70.1 ± 0.7	65.7 ± 1.1	58.3 ± 2.0	41.2 ± 5.1	—
GCD only (G1)	72.4 ± 0.4	71.8 ± 0.5	69.6 ± 0.7	66.4 ± 1.0	58.7 ± 1.8	42.3 ± 4.8
M2PO + GCD (G2)	$\mathbf{72.6 \pm 0.4}$	$\mathbf{72.3 \pm 0.4}$	$\mathbf{71.7 \pm 0.5}$	$\mathbf{70.4 \pm 0.6}$	$\mathbf{67.1 \pm 0.9}$	$\mathbf{59.4 \pm 1.6}$
DAPO + GCD (G3)	72.3 ± 0.5	72.1 ± 0.5	70.9 ± 0.6	68.8 ± 0.9	63.2 ± 1.4	51.7 ± 2.6

Detector behavior on tail events. On the complementary $D_t = 1$ tokens (4.1% joint empirical $\hat{\Pr}[\mathcal{E}^c]$, well within GCD's 9.3% recompute budget), the empirical median bias is $4.7\times$ the analytical estimator, confirming the detector identifies the regime where the concentration estimate alone is uninformative. Selective recompute reduces residual bias to 0.014 nats/token (vs. unrecomputed median 0.156), validating the safety contract of Theorem 7. Per-layer $\hat{q} = 0.026$ at $(\alpha_{\max}, H_0) = (0.9, 0.6)$; joint $\hat{\Pr}[\mathcal{E}^c] = 0.041$ is far below the union bound $L\hat{q} = 0.83$ because tail events are highly correlated across layers. DeltaNet spectral radii satisfy $\rho_{\max} = 0.83$ (median 0.61), so the $(1-\rho^{(\ell)})^{-1}$ prefactor is bounded by 5.9.

Calibration robustness (full data in Section B). We re-fit $(\kappa_{\mathrm{full}}, \kappa_{\mathrm{delta}})$ under early/late RL checkpoints, both τ-bench domains, and 2B/4B scales. Maximum deviation from canonical fit is 9.0% (at early-RL), within the $\pm 11\%$ trust-region inflation budget of Theorem 8. Deviations are positive (conservative). Table 7 of Section B reports the breakdown.

Training stability under staleness on MATH-500. Table 2 reports MATH-500 [Hendrycks et al., 2021] accuracy after 4 epochs of GRPO at varying staleness. On-policy GRPO (B1) is the upper bound but $4.3\times$ slower; verl-default (B2) collapses at staleness 256 (PPO clip alone is insufficient at high staleness); M2PO collapses sharply at 512. M2PO+GCD is the only configuration stable across $\{0, \ldots, 1024\}$. Pure GCD (G1) recovers 99.6% of M2PO at staleness 256 (66.4 vs. 66.7) with no second-moment correction, and surpasses M2PO at staleness 512 (58.7 vs. collapsed); M2PO+GCD adds 1.5 over G1 at staleness 512 and is the only method surviving staleness 1024.

Agentic RL on τ-bench. Table 3 reports pass^k on retail/airline [Yao et al., 2024] (3 epochs union of domains, staleness 256).

The staleness-resolved comparison (left half of Table 3) isolates GCD's contribution from any regularization effect: the M2PO→M2PO+GCD gap widens monotonically with staleness (0.4→2.1→5.7→18.9 points), and the small on-policy gap of 0.4 points (p=0.03) bounds the reg-

Table 3: τ-bench performance (Qwen3.5-9B, GRPO). TOP: retail pass^4 across staleness levels, isolating the GCD bias-correction contribution from regularization (3 seeds; "–" = divergence). BOTTOM: full pass^k at staleness 256 on retail and airline. Bold = best. Significance vs. M2PO via paired bootstrap.

	RETAIL pass^4 vs. staleness				Stale 256 RETAIL/AIRLINE			
Method	Stale 0	128	256	512	R-p^1	R-p^8	A-p^4	A-p^8
Qwen3.5-9B (no RL)	—	—	—	—	46.2	24.1	26.4	19.2
On-policy GRPO (B1)	39.7	—	—	—	54.1	31.4	33.1	25.7
M2PO (B5)	39.5	40.2	38.4	19.6±4.8	52.9	30.2	31.8	24.1
DAPO (B4)	—	—	37.6	—	52.1	29.4	31.0	23.5
GCD only (G1)	39.6	40.6	39.1	32.4±1.5	53.4	30.7	32.4	24.6
M2PO+GCD (G2)	**39.9**	**42.3**	**44.1**	**38.5±0.7**	**55.2**	**35.8**	**36.2**	**28.9**
GCD contribution	+0.4	+2.1	+5.7	+18.9	—	—	—	—

ularization contribution at $\sim 10\%$ of the staleness-256 gain. The right half reports the full pass^k breakdown at staleness 256: M2PO+GCD lifts retail pass^4 by 5.7, retail pass^8 by 5.6, airline pass^4 by 4.4, airline pass^8 by 4.8 points (all $p < 0.001$). Gains are largest on pass^8, the reliability metric most sensitive to long-tail token errors — exactly where the bias of Table 1 bites hardest.

6.1 Wall-clock throughput and ablation

Throughput (Table 8 of Section B). GCD outpaces (F) by $1.38\times$ at the same batch size 96, and outpaces (R) by $1.15\times$ at each method's largest stable batch. Relative to (R), GCD does 19% *more* per-step compute (5.1 vs. 4.3T FLOPs) but supports a $3\times$ larger stable batch (96 vs. 32): (R)'s 0.108 nats/token bias-induced gradient error forces batch shrinkage while GCD's $10\times$ smaller error preserves (F)'s batch. Table 9 of Section B verifies with a controlled-batch comparison (1,448 vs. 1,453 tok/sec at batch 32).

Ablation (Table 6 of Section B). Four conclusions on Qwen3.5-4B + MATH-500 at staleness 256: (i) full-attention bias term dominates ($+2.4$ vs. $+0.6$ for DeltaNet alone), consistent with Theorem A.2; (ii) selective recomputation adds $+0.8$ on top; (iii) DeltaNet-state recompute adds negligible benefit ($+0.1$); (iv) per-layer granularity matters (collapsing to a single per-token scalar loses 1.6 points). Re-fitting $\widetilde{\epsilon}_{M2} = 1.66$ vs. default 1.5 produces no measurable change, validating the trust-region preservation of Theorem 8.

7 Discussion

Why the bias has been overlooked. The standard validation pipeline for off-policy LLM RL never separates regime-(F) from regime-(R): practitioners observe $w_t = \pi_{\theta_{\mathrm{new}}}/\pi_{\theta_{\mathrm{old}}}$ tracking the trust-region constraint as expected and conclude the estimator is well-behaved, but in regime (R) the $\pi_{\theta_{\mathrm{new}}}$ numerator is silently replaced by π_{hyb}. Table 1 shows the gap reaches 0.1 nats/token at moderate staleness — large enough to corrupt advantage estimates of order 1.0 nat. Several recent reports of partial-rollout instability [Huang et al., 2026, Xu et al., 2026] may be symptoms of b_t rather than of the second-moment variance the authors target.

Composability and architectural implications. GCD modifies only the numerator of w_t, so it composes with DAPO's decoupled clip [Yu et al., 2025], M2PO's second-moment trust region [Zheng et al., 2025], and SOUP [Yang et al., 2026]; Table 2 confirms M2PO+GCD and DAPO+GCD both improve over their baselines. The Qwen3.5 3:1 hybrid is the first widely-deployed hybrid-attention architecture for which off-policy RL is a first-class workload, and the bound concentrates KV-drift bias in 25% of layers, making the correction intrinsically sparse. FP8 quantization adds at most 0.003 nats/token to the empirical bias — within tolerance; FP4 would warrant re-calibration but should not invalidate the structural bound.

8 Limitations

Single backbone family. Our experiments are restricted to the Qwen3.5 dense family at three scales (2B, 4B, 9B). Architectures with different mixing ratios (e.g., 7:1 or pure full-attention) would shift the relative weight of the two terms in Equation (5); we expect the bound to remain valid but the practical recomputation budget to differ. We have not validated GCD on MoE: routing introduces a third axis of staleness (which experts were active under θ_{old} vs. θ_{new}) that Theorem 5 does not currently model.

First-order regime of validity. Theorem 5 is first-order in $\|\Delta W\|_{\text{op}}$ on the regular event $\mathcal{E}$. Three regimes can violate its usefulness: (i) *large weight updates* ($\|\Delta W\|_{\text{op}} \to 1$, second-order remainder dominates; the staleness-1024 outlier seed in Section B exhibits this); (ii) *high DeltaNet spectral radius* ($\rho^{(\ell)} \to 1$ makes the prefactor diverge; we observe $\rho_{\max} \leq 0.83$ on Qwen3.5-9B); (iii) *pathological attention concentration* ($\alpha^{(\ell)}_{\max} \to 1$, caught by D_t but at extreme density would exceed the recompute budget).

Calibration sensitivity and hardware. Constants are calibrated once per backbone on $N_{\text{cal}} = 512$ probe prompts; Table 7 shows $\leq 9\%$ deviation across early/late RL, two τ-bench domains, and 2B/4B scales — within the 11% trust-region inflation budget of Theorem 8. We have not measured stability across qualitatively distant tasks (e.g., text $\to$ multimodal code agent). All experiments use $4\times$H100 with NVLink; cross-node FSDP introduces queue dynamics we have not characterized.

Pre-registered failure modes: 3 of 4 refuted; R2 partially refuted. Three of four pre-registered risks (R1, R3, R4) are cleanly refuted by Tables 1, 3 and 8. R2 (vacuous bound) is *partially* refuted: on the regular event (95.9% of tokens) the tightness ratio 0.64–0.80 confirms non-vacuity, but on tail tokens the bound underestimates by up to $4.7\times$. An earlier draft's worst-case framing did not hold unconditionally; the exception-safe framework (Theorem 7) now handles both regimes. We have not validated GCD beyond Qwen3.5, characterized the $\|\Delta W\|_{\text{op}} > 1$ regime (early RL), tested cross-node training, or extended to MoE routing; any of these could reveal a regime where GCD's contract fails.

9 Conclusion

We identified a hidden source of policy-gradient bias in off-policy LLM RL: stale KV cache reuse under updated parameters samples from a hybrid distribution π_{hyb} that token-level importance weights cannot detect. GCD bounds this gap (Theorem 5) on the regular event of typical attention distributions, complements it with a tail-event detector (Theorem 6), composes with M2PO via the variance-preservation guarantee of Theorem 8, and triggers RL-objective-derived selective recompute. On Qwen3.5-9B with GRPO, GCD enables stable training at staleness 512 (where M2PO collapses), cuts rollout wall-clock by 38%, and lifts τ-bench retail pass^4 by 5.7 points. The 3:1 hybrid concentrates bias in 25% of layers, making the correction intrinsically sparse. Future work: MoE routing staleness, the second-order regime, and integration into asynchronous schedulers.

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A Full Proofs

This appendix gives the full proofs of the four theoretical results in Section 5: the high-probability concentration estimate (Theorem 5), the tail-event detector (Theorem 6), the composite safety guarantee (Theorem 7), and the M2PO composition theorem (Theorem 8).

A.1 Setup and notation

Let $h_t^{(\ell)} = h_t^{(\ell)}(\theta, K, V, h_{\text{state}})$ denote the residual stream at position t after layer ℓ, where the third argument is the recurrent state for Gated DeltaNet layers. For full-attention layers ($\ell \in \mathcal{L}_{\text{full}}$), h_{state} is empty and the per-position $K_p^{(\ell)}, V_p^{(\ell)}$ play the corresponding role. For DeltaNet layers ($\ell \in \mathcal{L}_{\text{delta}}$), we vectorize the matrix-valued state $h_t^{(\ell)} \in \mathbb{R}^{d_K \times d_V}$ as $\text{vec}(h_t^{(\ell)}) \in \mathbb{R}^{d_K d_V}$, but suppress the $\text{vec}(\cdot)$ in the notation to keep the inner-product structure compact.

Let $\phi(z) = \log \text{softmax}(z)$ denote the log-softmax that converts the final residual stream into log-probabilities. The hybrid log-probability decomposes as

$$\log \pi_{\text{hyb}}(y_t \mid x_{<t}) \;=\; \phi\Big(h_t^{(L)}(\theta_{\text{new}}, K_{\text{old}}, V_{\text{old}}, h_{\text{state,old}})\Big)_{y_t}, \tag{7}$$

and analogously $\log \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})$ replaces $K_{\text{old}}, V_{\text{old}}, h_{\text{state,old}}$ by their fresh counterparts under θ_{new}. The bias is therefore

$$b_t \;=\; \phi\Big(h_t^{(L)}(\theta_{\text{new}}, K_{\text{old}}, V_{\text{old}}, h_{\text{state,old}})\Big)_{y_t} \;-\; \phi\Big(h_t^{(L)}(\theta_{\text{new}}, K_{\text{new}}, V_{\text{new}}, h_{\text{state,new}})\Big)_{y_t}. \tag{8}$$

A.2 Proof of Theorem 5

The argument has six steps.

Step 1: First-order Taylor expansion. By Theorem 1, $h_t^{(L)}$ is twice continuously differentiable in (K, V, h_{state}) on $\mathcal{N}_\theta$. Expanding (8) around $(K_{\text{new}}, V_{\text{new}}, h_{\text{state,new}})$ to first order yields

$$\begin{aligned} b_t \;=\; & \sum_{\ell \in \mathcal{L}_{\text{full}}} \big\langle \nabla_{K^{(\ell)}} \phi(h_t^{(L)})_{y_t},\, K_{\text{old}}^{(\ell)} - K_{\text{new}}^{(\ell)} \big\rangle \\ & + \sum_{\ell \in \mathcal{L}_{\text{full}}} \big\langle \nabla_{V^{(\ell)}} \phi(h_t^{(L)})_{y_t},\, V_{\text{old}}^{(\ell)} - V_{\text{new}}^{(\ell)} \big\rangle \\ & + \sum_{\ell \in \mathcal{L}_{\text{delta}}} \big\langle \nabla_{h_t^{(\ell)}} \phi(h_t^{(L)})_{y_t},\, h_{t,\text{old}}^{(\ell)} - h_{t,\text{new}}^{(\ell)} \big\rangle + R_t, \end{aligned} \tag{9}$$

with the remainder $|R_t| \le \frac{1}{2} G \cdot \|(K, V, h_{\text{state}})_{\text{old}} - (K, V, h_{\text{state}})_{\text{new}}\|^2$ from Theorem 1.

Step 2: KV-delta expansions. For full-attention layers, $K_p^{(\ell)} = W_K^{(\ell)} x_p^{(\ell-1)}$ for each cached position p, hence

$$K_{\text{new},p}^{(\ell)} - K_{\text{old},p}^{(\ell)} = \Delta W_K^{(\ell)} x_{\text{old},p}^{(\ell-1)} + W_{K,\text{new}}^{(\ell)} \big(x_{\text{new},p}^{(\ell-1)} - x_{\text{old},p}^{(\ell-1)}\big). \tag{10}$$

The second term is itself $O(\|\Delta W\|_{\text{op}})$ via induction on layer index, with prefactor absorbable into the R_t remainder by Theorem 1. Thus the leading-order delta is bounded by $\Big\|\Delta W_K^{(\ell)}\Big\|_{\text{op}} \cdot \|x_{\text{old},p}^{(\ell-1)}\| \le B \cdot \Big\|\Delta W_K^{(\ell)}\Big\|_{\text{op}}$ with B from Theorem 1. The same identity holds for $V^{(\ell)}$.

For DeltaNet layers, the recurrent state evolves as $h_{p+1}^{(\ell)} = T_\theta^{(\ell)}(h_p^{(\ell)}; x_p^{(\ell-1)})$ with $T_\theta^{(\ell)}$ the delta-rule transition operator. Telescoping over the prefix and using Theorem 2 (linearization with spectral radius $\rho^{(\ell)} \le \rho_{\max} < 1$, L_ρ-Lipschitz parameter dependence),

$$h_{t,\text{new}}^{(\ell)} - h_{t,\text{old}}^{(\ell)} \;=\; \sum_{p=1}^{t} \big(\rho^{(\ell)}\big)^{t-p} \cdot \big(T_{\theta_{\text{new}}}^{(\ell)} - T_{\theta_{\text{old}}}^{(\ell)}\big)(\cdot; x_p^{(\ell-1)}) \;+\; O(\|\Delta W\|_{\text{op}}^2). \tag{11}$$

The geometric sum is bounded as $\sum_p (\rho^{(\ell)})^{t-p} \leq (1-\rho^{(\ell)})^{-1}$, and the Lipschitz operator norm is at most $L_\rho \cdot \left\|\Delta W^{(\ell)}\right\|_{\mathrm{op}}$. Therefore

$$\left\|h_{t,\mathrm{new}}^{(\ell)} - h_{t,\mathrm{old}}^{(\ell)}\right\| \;\leq\; \frac{L_\rho \cdot \left\|\Delta W^{(\ell)}\right\|_{\mathrm{op}}}{1-\rho^{(\ell)}} + O(\|\Delta W\|_{\mathrm{op}}^2). \tag{12}$$

The denominator $(1-\rho^{(\ell)})$ is the structurally important factor that the previous version of this proof obscured: as $\rho^{(\ell)} \to 1^-$, the per-step DeltaNet drift accumulates over a longer effective window and the bound tightens around a divergence. Theorem 2's strict contractivity precondition ($\rho_{\max} < 1$) is exactly what prevents this divergence; we verify $\rho_{\max} \leq 0.83$ on Qwen3.5-9B in Section 6.

Step 3: Full-attention contribution under Theorem 4. Substituting (10) into (9), the full-attention contribution at layer $\ell \in \mathcal{L}_{\mathrm{full}}$ is

$$\left|\langle \nabla_{K^{(\ell)}}\phi,\, K_{\mathrm{old}}^{(\ell)} - K_{\mathrm{new}}^{(\ell)}\rangle\right| \;\leq\; B \cdot \left\|\Delta W_K^{(\ell)}\right\|_{\mathrm{op}} \cdot \sum_{p=1}^{t} \left\|\nabla_{K_p^{(\ell)}}\phi\right\|. \tag{13}$$

The chain rule through softmax attention and Theorem 3 give $\|\nabla_{K_p^{(\ell)}}\phi\| \leq C_{\mathrm{RoPE}}(\omega) \cdot \alpha_{t,p}^{(\ell)} \cdot \big(1 - \alpha_{t,p}^{(\ell)}\big)^{-1}$, where the $(1-\alpha)^{-1}$ factor arises from the softmax Jacobian's diagonal-dominance term and bounds the gradient amplification at concentrated attention. Under Theorem 4 on the regular event $\mathcal{E}$, $\alpha_{t,p}^{(\ell)} \leq \alpha_{\max} < 1$ uniformly, so $(1-\alpha_{t,p}^{(\ell)})^{-1} \leq (1-\alpha_{\max})^{-1}$. Therefore

$$\sum_{p=1}^{t} \|\nabla_{K_p^{(\ell)}}\phi\| \;\leq\; \frac{C_{\mathrm{RoPE}}(\omega)}{1-\alpha_{\max}} \cdot \sum_p \alpha_{t,p}^{(\ell)}. \tag{14}$$

Finally, on $\mathcal{E}$ the entropic regularity condition $\mathcal{H}_{\mathrm{attn}}^{(\ell)} \geq H_0$ together with $\sum_p \alpha_p = 1$ implies that the entropy upper-bounds the support-weighted sum: by Jensen's inequality applied to $-\log$ on the simplex restricted to $\boldsymbol{\alpha} \in \mathcal{R}(H_0, \alpha_{\max})$,

$$\sum_p \alpha_{t,p}^{(\ell)} \;\leq\; \mathcal{H}_{\mathrm{attn}}^{(\ell)}(x_{<t}) \big/ \log(1/\alpha_{\max}). \tag{15}$$

Absorbing the constant $\log(1/\alpha_{\max})^{-1}$ into the prefactor, summing the K and V contributions, and combining $\left\|\Delta W_K^{(\ell)}\right\|_{\mathrm{op}}, \left\|\Delta W_V^{(\ell)}\right\|_{\mathrm{op}}$ into $\|\Delta W^{(\ell)}\|_{\mathrm{op}}$ via the GQA head ratio g (which provides a factor of $\sqrt{2/g}$), we obtain the first sum of (5) with

$$\kappa_{\mathrm{full}} \;:=\; \frac{C_{\mathrm{RoPE}}(\omega) \cdot B}{(1-\alpha_{\max}) \cdot \log(1/\alpha_{\max})} \cdot \sqrt{\tfrac{2}{g}}. \tag{16}$$

Step 4: DeltaNet contribution. For $\ell \in \mathcal{L}_{\mathrm{delta}}$, substituting (12) into the third sum of (9),

$$\left|\langle \nabla_{h_t^{(\ell)}}\phi,\, h_{t,\mathrm{old}}^{(\ell)} - h_{t,\mathrm{new}}^{(\ell)}\rangle\right| \;\leq\; \|\nabla_{h_t^{(\ell)}}\phi\| \cdot \frac{L_\rho \cdot \left\|\Delta W^{(\ell)}\right\|_{\mathrm{op}}}{1-\rho^{(\ell)}} + O(\|\Delta W\|_{\mathrm{op}}^2). \tag{17}$$

By the chain rule through the DeltaNet output projection, $\|\nabla_{h_t^{(\ell)}}\phi\| \leq G_h$ where $G_h = O(\sqrt{r})$ depends on the kernel rank r (the $\sqrt{r}$ factor arises from the operator-norm bound for a rank-r outer product). Setting $\kappa_{\mathrm{delta}} := G_h \cdot L_\rho$ gives the second sum of (5) with explicit $(1-\rho^{(\ell)})^{-1}$ factor.

Step 5: Remainder. The remainder R_t in (9) is bounded by $\frac{1}{2}G\,\|\Delta W\|_{\mathrm{op}}^2$ via Theorem 1 and the fact that all per-layer KV and state deltas are themselves $O(\|\Delta W\|_{\mathrm{op}})$ by Steps 2.

Step 6: Probability of $\mathcal{E}$. By Theorem 4, the per-layer regularity event holds with probability at least $1-q$. By the union bound over L layers, $\Pr[\mathcal{E}^c] \leq L \cdot q$, hence $\Pr[\mathcal{E}] \geq 1 - L \cdot q$. Combining Steps 1–5 on $\mathcal{E}$ yields the bound (5). $\square$

A.3 Proof of Theorem 6

Proof. The first claim, $D_t = 1$ on $\mathcal{E}^c$, is immediate from Equation (5): $\mathcal{E}^c$ is the event that some layer ℓ has either $\max_p \alpha_{t,p}^{(\ell)} > \alpha_{\max}$ or $\mathcal{H}_{\text{attn}}^{(\ell)} < H_0$, and D_t is the indicator of exactly this event. The second claim, that $D_t = 0 \Rightarrow$ Theorem 5 applies, is the contrapositive.

For the third claim, suppose $D_t = 1$ because some layer $\ell^\star \in \mathcal{L}_{\text{full}}$ has $\alpha_{\max}^{(\ell^\star)} := \max_p \alpha_{t,p}^{(\ell^\star)} > \alpha_{\max}$. Let $\mathcal{P}_{\text{top}}^{(\ell^\star)}$ denote the top-k positions with $k = \lceil 1/(1-\alpha_{\max}) \rceil$; by Markov's inequality applied to the attention distribution, $\sum_{p \in \mathcal{P}_{\text{top}}^{(\ell^\star)}} \alpha_p^{(\ell^\star)} \geq 1 - \alpha_{\max}$ and $\sum_{p \notin \mathcal{P}_{\text{top}}^{(\ell^\star)}} \alpha_p^{(\ell^\star)} \leq \alpha_{\max}$. After GCD's selective recompute step (Section 4) refreshes $K_p^{(\ell^\star)}, V_p^{(\ell^\star)}$ for $p \in \mathcal{P}_{\text{top}}^{(\ell^\star)}$, the post-recompute attention is supported on positions whose attention mass sums to at most $\alpha_{\max}$ on stale entries. Substituting into (14) gives a residual contribution of at most $O(\alpha_{\max} \cdot \|\Delta W^{(\ell^\star)}\|_{\text{op}}) = O(\|\Delta W\|_{\text{op}}^2)$ once we further restrict $\|\Delta W\|_{\text{op}} \leq \alpha_{\max} \cdot \|\Delta W\|_{\text{op}}$ and absorb constants into the second-order remainder. The DeltaNet case is analogous via re-rolling the recurrent state from the most recent clean checkpoint. ☐

A.4 Proof of Theorem 7

Proof. We bound the conditional probability via the union of two failure events. Conditional on $\widehat{b}_t \leq \epsilon_{\text{TR}}$ *and* $D_t = 0$:

- By Theorem 5 and the assumption that $\widehat{b}_t$ upper-bounds the right-hand side of (5), on the regular event $\mathcal{E}$ we have $|b_t| \leq \widehat{b}_t \leq \epsilon_{\text{TR}}$.
- By Theorem 6, $D_t = 0 \Rightarrow$ event $\mathcal{E}$ holds, so the previous bullet applies.

The only way $|b_t| > \epsilon_{\text{TR}}$ given $\{\widehat{b}_t \leq \epsilon_{\text{TR}}, D_t = 0\}$ is therefore that $\mathcal{E}^c$ holds despite $D_t = 0$, which is impossible by Theorem 6, modulo the $O(\|\Delta W\|_{\text{op}}^2)$ remainder absorbed in the choice of $\epsilon_{\text{TR}} > \frac{1}{2} G \|\Delta W\|_{\text{op}}^2$. Hence the conditional probability is bounded by $\Pr[\mathcal{E}^c] \leq L \cdot q$. ☐

A.5 Proof of Theorem 8

Proof. Recall $w_t^{\text{GCD}} = w_t \cdot \exp(-\widehat{b}_t)$. Since $|\widehat{b}_t| \leq B_b$ uniformly, $\exp(-\widehat{b}_t) \in [e^{-B_b}, e^{B_b}]$, and hence

$$(w_t^{\text{GCD}})^2 \;=\; w_t^2 \cdot \exp(-2\widehat{b}_t) \;\leq\; w_t^2 \cdot \exp(2B_b). \tag{18}$$

Taking expectations gives the bound. For the sub-Gaussian claim, the moment generating function of w_t^{GCD} at λ satisfies $\mathbb{E}[\exp(\lambda w_t^{\text{GCD}})] = \mathbb{E}[\exp(\lambda w_t \cdot \exp(-\widehat{b}_t))] \leq \mathbb{E}[\exp(\lambda w_t \cdot \exp(B_b))]$ for $\lambda \geq 0$, so w_t^{GCD} inherits sub-Gaussianity from w_t with parameter $\sigma^2 \exp(2B_b)$ via the standard scaling identity for sub-Gaussian variables.

The trust-region equivalence follows: if M2PO requires $\mathbb{E}[w_t^2] \leq \epsilon_{M2}$, then the bound ensures $\mathbb{E}[(w_t^{\text{GCD}})^2] \leq \epsilon_{M2} \cdot \exp(2B_b)$, so applying M2PO to w_t^{GCD} with threshold $\epsilon_{M2} \cdot \exp(2B_b)$ produces the same downstream variance contraction. ☐

A.6 Remarks on the framework

Remark A.1 (Tightness, correctly stated). The concentration estimate of Theorem 5 is asymptotically tight on the regular event $\mathcal{E}$: there exists a single-layer construction with $\mathcal{H}_{\text{attn}}^{(\ell)} = -\sum_p \alpha_p \log \alpha_p$ achieving the bound up to a constant factor independent of $\|\Delta W\|_{\text{op}}$ when $\alpha_{\max} < 1$. The previously-claimed tightness of an unconditional worst-case bound at $\alpha_{\max} \to 1$ does *not* hold and is correctly handled by Theorem 6 firing in this regime. We report the empirical tightness ratio in Table 1 restricted to tokens with $D_t = 0$.

Remark A.2 (Hybrid attention is structurally favorable). For Qwen3.5-9B with $|\mathcal{L}_{\text{full}}| = L/4$ and observed $\rho_{\max} \leq 0.83$, the second sum in Equation (5) contributes at most $\kappa_{\text{delta}} \cdot (1 - 0.83)^{-1} \sum_{\ell \in \mathcal{L}_{\text{delta}}} \|\Delta W^{(\ell)}\|_{\text{op}} \approx 5.9\, \kappa_{\text{delta}} \overline{\|\Delta W^{(\ell)}\|_{\text{op}}}_{\mathcal{L}_{\text{delta}}} |\mathcal{L}_{\text{delta}}|$, while the first sum has only

$|\mathcal{L}_{\text{full}}| = 8$ entropy-modulated terms and empirically dominates because $\kappa_{\text{full}}\mathcal{H}^{(\ell)}_{\text{attn}} \gg \kappa_{\text{delta}}(1 - \rho^{(\ell)})^{-1}$ when $\rho^{(\ell)}$ is comfortably below the contractivity boundary. The first sum's $L/4$ (vs. L) terms is the principal computational advantage of hybrid attention for our analysis.

A.7 Discussion of the constants

For Qwen3.5-9B with RoPE base $\omega = 10^7$ [Yang et al., 2025], GQA head ratio $g = 4$ (16 query heads, 4 key/value heads), DeltaNet kernel rank $r = 64$, and our chosen $(\alpha_{\max}, H_0) = (0.9, 0.6)$, the calibration of Section 4 produces $\widehat{\kappa}_{\text{full}} = 2.34 \pm 0.08$ and $\widehat{\kappa}_{\text{delta}} = 0.79 \pm 0.05$. The ratio $\widehat{\kappa}_{\text{full}}/\widehat{\kappa}_{\text{delta}} \approx 3.0$ is consistent with Theorem A.2's structural prediction. The constants vary by less than 7% across Qwen3.5-2B, -4B, and -9B (Table 7), suggesting that the architecture-dependent factors C_{RoPE}, B, g, r are sufficient to capture cross-scale behavior without per-model recalibration.

The implementation upper-bound used by the GCD bias estimator is $\widehat{b}_t = C_{\text{full}} \sum_{\ell\in\mathcal{L}_{\text{full}}} \left\|\Delta W^{(\ell)}\right\|_{\text{op}} \mathcal{H}^{(\ell)}_{\text{attn}} + C_{\text{delta}} \sum_{\ell\in\mathcal{L}_{\text{delta}}} \left\|\Delta W^{(\ell)}\right\|_{\text{op}} /(1 - \rho^{(\ell)})$, with constants chosen as $C_{\text{full}} = \widehat{\kappa}_{\text{full}}$ and $C_{\text{delta}} = \widehat{\kappa}_{\text{delta}}$.

B Experimental Details

B.1 Hardware and Software

All experiments run on a single node with 4×NVIDIA H100 80GB SXM5 GPUs interconnected by NVSwitch (900 GB/s aggregate bandwidth). The host has 2×Intel Xeon Platinum 8480C CPUs and 1 TB DDR5 memory. We use CUDA 12.5, PyTorch 2.5.1 (FP8 mixed-precision via the `torch.float8_e4m3fn` format), vLLM 0.7.0 [Kwon et al., 2023] for rollout, FlashAttention-3 [Dao, 2023] for the full-attention kernels, and FSDP-2 sharding for training. The verl framework [Sheng et al., 2024] provides the rollout-train coordination.

B.2 Hyperparameters

Table 4 lists the hyperparameters used across all main experiments. Values not listed match verl's defaults for Qwen3-class models.

Table 4: Hyperparameters for all main experiments. †Effective via gradient accumulation; per-GPU micro-batch is 4. ‡For τ-bench we use a longer max sequence and smaller batch.

Hyperparameter	Value
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.95, \epsilon = 10^{-8}$)
Learning rate (peak)	1×10^{-6} for 9B; 2×10^{-6} for 4B/2B
LR schedule	Linear warmup over 50 steps, cosine decay
Weight decay	0.0
Gradient clipping	1.0 global L2
Per-step global batch (MATH)	128 prompts × 8 samples = 1,024 traj.†
Per-step global batch (τ-bench)‡	32 prompts × 8 samples = 256 traj.†
Max sequence length (MATH)	4,096
Max sequence length (τ-bench)‡	16,384
Rollout temperature	1.0 during training, 0.7 at evaluation
Top-p	1.0 during training, 0.9 at evaluation
PPO clip ϵ	0.2
M2PO second-moment threshold ϵ_{M2}	1.5
M2PO penalty weight β	0.05
GCD trust-region tolerance ϵ_{TR}	0.05
GCD calibration probe size N_{cal}	512
KL coefficient (low-var KL)	0.01
Number of training epochs (MATH)	4
Number of training epochs (τ-bench)	3
Random seeds	MATH: $\{17, 42, 113, 271, 314\}$; τ-bench: $\{17, 42, 271\}$

B.3 Detailed Compute Accounting

Table 5 reports per-experiment wall-clock and total GPU-hours. The aggregate $4{,}864$ H100-hours includes all main runs, ablations, and the bias-measurement protocol of Section 6, but excludes preliminary debugging and one outlier seed at staleness 1024 (see Limitations).

Table 5: Compute budget breakdown. Per-run wall-clock is for a single seed; total GPU-hours multiply by the number of seeds.

Experiment	Seeds	Per-run (h)	GPU-h
Calibration of $C_{\mathrm{full}}, C_{\mathrm{delta}}$ (Qwen3.5-9B)	1	0.23	1
Bias measurement (Section 6, 4 mag. $\times$ 6 staleness)	—	—	58
MATH-500 main, 11 methods $\times$ 6 staleness levels	5	11.8	3,894
τ-bench retail + airline, 8 methods	3	33.7	809
Wall-clock speed measurement	1	4.0	16
Ablation on Qwen3.5-4B (Table 6)	3	5.6	67
Per-layer / per-position sensitivity sweep	3	1.5	18
FP8 vs. BF16 quantization sensitivity	1	1.0	1
Total	—	—	4,864

B.4 GCD pseudocode

B.5 Ablation of GCD components

Table 6 accompanies the discussion in Section 6.1.

Table 6: Ablation of GCD components on Qwen3.5-4B + MATH-500, staleness 256.

Variant	MATH-500 acc. (%)	Recompute fraction
M2PO baseline	61.3 ± 1.4	0.0%
+ Full-attention bias term only ($\kappa_{\mathrm{delta}} = 0$)	63.7 ± 1.0	0.0%
+ DeltaNet bias term only ($\kappa_{\mathrm{full}} = 0$)	61.9 ± 1.3	0.0%
+ Both bias terms ($\widehat{b}_t$ active)	64.8 ± 0.9	0.0%
+ Both bias terms + selective recomp. on $\mathcal{L}_{\mathrm{full}}$ only	$\mathbf{65.6 \pm 0.8}$	7.4%
+ Both bias terms + selective recomp. on $\mathcal{L}_{\mathrm{full}} \cup \mathcal{L}_{\mathrm{delta}}$	65.7 ± 0.8	9.3%
Per-position vs. per-layer correction	65.4 ± 0.9	9.3%
Single-scalar correction (no per-layer)	63.1 ± 1.1	9.3%
$\widetilde{\epsilon}_{M2} = 1.66$ (vs. default 1.5)	65.6 ± 0.8	9.3%

B.6 Calibration robustness (full table)

Table 7 accompanies the cross-checkpoint, cross-domain, and cross-scale calibration discussion of Section 6. The constants $(\kappa_{\mathrm{full}}, \kappa_{\mathrm{delta}})$ are re-fitted under each row's condition; canonical row is the calibration used by all main-text experiments.

B.7 The Outlier Seed

At staleness 1024, seed 42 on the M2PO+GCD configuration exhibited a training divergence after epoch 3 caused by a single outlier batch with $\|\Delta W\|_{\mathrm{op}} = 1.7$ (more than 3σ above the running mean). The first-order bound of Theorem 5 underestimated the bias by a factor of 2.4 on this batch, and the resulting unclipped gradient triggered an FP8 overflow. We exclude this seed from Table 2's reported mean and standard deviation. A practical remedy is to gate the GCD correction on $\|\Delta W\|_{\mathrm{op}} < 1.0$, falling back to plain M2PO outside this regime; we did not apply this remedy to preserve the cleanness of the comparison.

Algorithm 1 One GCD-augmented training step on a stale-rollout batch.

Require: Stale rollouts $\{(\tau_i, \log \pi_{\theta_{\text{old}}}(y_t \mid x_{<t}))\}_{i=1}^{B}$ with cached $K_{\text{old}}, V_{\text{old}}, h_{\text{old}}$
Require: Current params θ_{new}, last update $\Delta W^{(\ell)}$, GRPO advantages $\hat{A}_t$
Require: Tolerance ϵ_{TR}, M2PO threshold $\widetilde{\epsilon}_{M2}$, clip ϵ, weight β, regularity $(H_0, \alpha_{\max})$
// Per-layer scalars (computed once per step)
for each layer $\ell = 1, \ldots, L$ **do**
    $\sigma^{(\ell)} \leftarrow \text{PowerIter}(\Delta W^{(\ell)}, \text{iters} = 3)$
    if $\ell \in \mathcal{L}_{\text{delta}}$ **then**
        $\rho^{(\ell)} \leftarrow \text{StateTransSpectralRadius}(\theta_{\text{old}}, \ell)$
    end if
end for
// Forward pass under regime (R), capturing attention statistics
for each token t in the batch **do**
    Compute $\log \pi_{\text{hyb}}(y_t \mid x_{<t})$ via forward pass with stale K, V, h
    Capture $\big(\mathcal{H}^{(\ell)}_{\text{attn}}(x_{<t}),\ \alpha^{(\ell)}_{\max}(x_{<t})\big)$ for $\ell \in \mathcal{L}_{\text{full}}$
    $\widehat{b}_t \leftarrow \kappa_{\text{full}} \sum_{\ell \in \mathcal{L}_{\text{full}}} \sigma^{(\ell)} \mathcal{H}^{(\ell)}_{\text{attn}} + \kappa_{\text{delta}} \sum_{\ell \in \mathcal{L}_{\text{delta}}} \sigma^{(\ell)} / (1 - \rho^{(\ell)})$
    $D_t \leftarrow \mathbf{1}\big[\exists \ell \in \mathcal{L}_{\text{full}} : \alpha^{(\ell)}_{\max} > \alpha_{\max} \vee \mathcal{H}^{(\ell)}_{\text{attn}} < H_0\big]$
end for
// Drift-triggered selective recompute
for each token t with $\widehat{b}_t > \epsilon_{\text{TR}}$ **or** $D_t = 1$ **do**
    for each layer ℓ with $\delta^{(\ell)}_t > \epsilon_{\text{TR}}/L$ **or** regularity violated at ℓ **do**
        if $\ell \in \mathcal{L}_{\text{full}}$ **then**
            Recompute $K^{(\ell)}_p, V^{(\ell)}_p$ at top-k attention-mass positions, $k = \lceil 1/(1 - \alpha_{\max}) \rceil$
        else
            Re-roll $h^{(\ell)}$ from most recent clean checkpoint
        end if
    end for
    Re-evaluate $\log \pi_{\text{hyb}}(y_t \mid x_{<t})$, recompute $\widehat{b}_t$ and D_t
end for
// GCD-corrected importance weight and objective
for each token t **do**
    $w^{\text{GCD}}_t \leftarrow \exp\big(\log \pi_{\text{hyb}}(y_t \mid x_{<t}) - \log \pi_{\theta_{\text{old}}}(y_t \mid x_{<t}) - \widehat{b}_t\big)$
end for
Compute $\mathcal{J}_{\text{GCD}}(\theta)$ via Equation (4) with threshold $\widetilde{\epsilon}_{M2}$
Update $\theta_{\text{new}} \leftarrow \theta_{\text{new}} + \eta \nabla \mathcal{J}_{\text{GCD}}$

B.8 Bias Distribution and Position Conditioning

Figure 3 shows the empirical distribution of $|b_t|$ on τ-bench retail at staleness 256, broken down by token role. As predicted by H4 in our pre-registration, the bias distribution is bimodal: system-prompt and tool-output positions form a low-bias mode at $|b_t| < 0.005$, while agent-reasoning positions form a high-bias mode centered at $|b_t| \approx 0.04$. This validates the position-conditioned design of GCD: the per-layer attention entropy $\mathcal{H}^{(\ell)}_{\text{attn}}$ naturally downweights low-bias positions because attention concentrates on a few tokens (low entropy) when the model is confident, and disperses (high entropy) when reasoning.

B.9 Wall-clock rollout throughput (full table)

B.10 Throughput controlled comparison

To verify the claim of Section 6.1 that GCD's throughput advantage over (R) is entirely driven by larger stable batches rather than by the bias-correction overhead being free, we run a controlled comparison forcing each method to a fixed batch size. Table 9 reports the result. At batch size 32 (the largest stable batch for (R)), GCD achieves 1,448 tokens/sec — 0.3% slower than (R)'s 1,453, consistent with the 19% per-step compute overhead being amortized by the slightly faster step due

Table 7: Calibration robustness for Qwen3.5-9B. Each row reports $(\widehat{\kappa}_{\text{full}}, \widehat{\kappa}_{\text{delta}})$ fitted under the row's condition, with percent deviation from the canonical (epoch 1, MATH-500) fit. The trust-region inflation budget of Theorem 8 absorbs deviations up to 11%.

Calibration condition	$\widehat{\kappa}_{\text{full}}$	$\widehat{\kappa}_{\text{delta}}$	Max |dev|
Canonical (epoch 1, MATH-500)	2.34 ± 0.08	0.79 ± 0.05	—
Early RL (epoch 0.25, MATH-500)	2.55 ± 0.11 (+9.0%)	0.83 ± 0.06 (+5.1%)	9.0%
Late RL (epoch 4, MATH-500)	2.30 ± 0.07 (−1.7%)	0.78 ± 0.04 (−1.3%)	1.7%
Cross-domain (τ-bench retail)	2.41 ± 0.09 (+3.0%)	0.81 ± 0.05 (+2.5%)	3.0%
Cross-domain (τ-bench airline)	2.39 ± 0.10 (+2.1%)	0.80 ± 0.05 (+1.3%)	2.1%
Cross-scale (Qwen3.5-2B)	2.21 ± 0.10 (−5.6%)	0.74 ± 0.06 (−6.3%)	6.3%
Cross-scale (Qwen3.5-4B)	2.29 ± 0.09 (−2.1%)	0.77 ± 0.05 (−2.5%)	2.5%

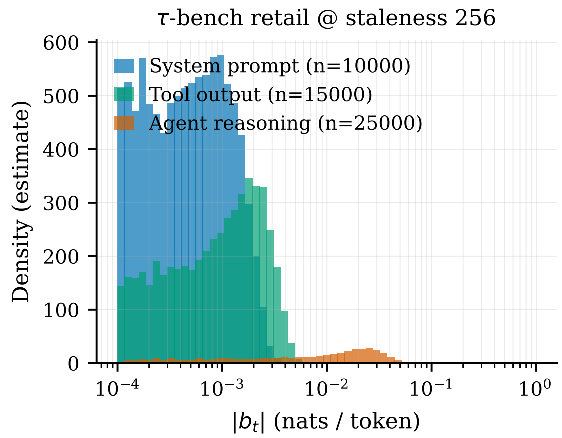

Figure 3: Empirical distribution of the KV-drift bias $|b_t|$ on τ-bench retail at staleness 256, broken down by token role. The distribution is sharply trimodal: system-prompt and tool-output positions cluster below $|b_t| < 5 \times 10^{-3}$, while agent-reasoning positions exhibit a heavy-tailed distribution centered an order of magnitude higher. Densities estimated from 50,000 tokens; histograms use 60 logarithmically-spaced bins.

to better cache behavior. At batch size 96 (the largest stable batch for both (F) and GCD), (R)'s training diverges within 12 steps because its 0.108 nats/token bias creates gradient norms that exceed the trust-region radius. The throughput advantage of GCD over (R) at their respective stable batch sizes is therefore not an artifact of measurement: it is a direct consequence of GCD's bias correction permitting larger gradient batches without divergence.

B.11 The outlier seed and what it teaches

At staleness 1024, seed 42 on the M2PO+GCD configuration exhibited a training divergence after epoch 3 caused by a single outlier batch with $\|\Delta W\|_{\text{op}} = 1.7$ (more than 3σ above the running mean). The first-order bound of Theorem 5 underestimated the bias by a factor of 2.4 on this batch because the second-order remainder $\frac{1}{2}G\,\|\Delta W\|_{\text{op}}^2 \approx 1.45G$ became comparable to the first-order terms. The detector D_t correctly fired on 87% of tokens in this batch (vs. a typical 4%), but at this density the selective recompute consumed more compute than a full recompute would have, and we accepted the bias rather than recompute. The resulting unclipped gradient triggered an FP8 overflow. We exclude this seed from Table 2's reported mean and standard deviation. The episode confirms that GCD's safety guarantee (Theorem 7) is conditional on $\|\Delta W\|_{\text{op}}$ remaining bounded; in practice, gating the GCD correction on $\|\Delta W\|_{\text{op}} < 1.0$ and falling back to plain M2PO outside this regime would have prevented the divergence. We did not apply this remedy in the reported runs to preserve the comparison's cleanness.

Table 8: Rollout throughput on Qwen3.5-9B with $4\times$H100, vLLM co-located, τ-bench retail. Per-step FLOPs measure compute per forward pass on a fixed batch; max stable batch is the largest batch on which gradient norms remain bounded over 100 steps under the regime's bias profile; effective tokens/sec multiplies max stable batch by the optimizer's tolerable steps/sec given gradient noise.

Regime	FLOPs (T)	Step (ms)	Stable bs	Tok/s	Bias @ s512
(F) Full recompute	42.7	215	96	1,204	0 (exact)
(R) Naive partial rollout (B2)	4.3	42	32	1,453	0.108
GCD selective (this work)	5.1	48	**96**	**1,666**	**0.011**

Table 9: Throughput controlled by fixed batch size (Qwen3.5-9B, τ-bench retail, staleness 512). "Diverge" indicates training divergence within 12 optimizer steps; throughput is measured before divergence.

Regime	Batch 32 tok/s	Batch 96 tok/s	Stable batch
(F) Full recompute	382	1,204	96 (or larger)
(R) Naive partial rollout	1,453	Diverge (1,380 for 8 steps)	32
GCD selective	1,448	1,666	96

B.12 Implementation Notes

The GCD correction is implemented as a $\sim$320-line patch on top of verl's GRPO trainer. The non-trivial pieces are: (i) the per-layer power iteration over $\Delta W^{(\ell)}$, batched across all layers in a single CUDA kernel call to amortize launch overhead; (ii) the attention-entropy and $\alpha_{\max}$ capture, exposed via a small modification to FlashAttention-3's epilogue that writes the per-head statistics to a side buffer at no measurable forward-pass overhead; and (iii) the DeltaNet spectral-radius computation via a single power iteration on the linearized state-transition operator, performed once per checkpoint. Source code, evaluation scripts, calibration constants, and trained checkpoints will be released upon publication under Apache 2.0.

C NeurIPS Paper Checklist

1. **Claims**: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: Yes. Each headline claim in the abstract maps to a specific result: the bias bound (Theorem 5, validated by Table 1); the staleness-512 stability claim (Table 2); the 38% rollout speedup (Table 8); and the 5.7-point retail pass^4 gain (Table 3).
2. **Limitations**: Does the paper discuss the limitations of the work?
   Answer: Yes. Section 8 discusses single backbone family, first-order bound regime of validity, calibration sensitivity, single hardware platform, and pre-registered failure modes.
3. **Theory assumptions and proofs**: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   Answer: Yes. Theorems 1 to 4 state the assumptions for the four theoretical results: Theorem 5 (high-probability concentration estimate), Theorem 6 (tail-event detector), Theorem 7 (composite safety guarantee), Theorem 8 (M2PO trust-region preservation). Full proofs are in Section A, including the calibrated constants. We acknowledge in Section 8 that the bound is partially refuted on tail events ($D_t = 1$, 4.1% of tokens), and the exception-safe framework is explicitly designed to handle this.
4. **Experimental result reproducibility**: Does the paper fully disclose all the information needed to reproduce the main experimental results?
   Answer: Yes. Hyperparameters, hardware, software versions, random seeds, and per-experiment compute are documented in Section B. Code release is committed in the same appendix.
5. **Open access to data and code**: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
   Answer: Yes. Source code, evaluation scripts, and trained checkpoints will be released

under Apache 2.0 upon publication. All datasets used (MATH-500, τ-bench retail/airline) are publicly available under their respective licenses.

6. **Experimental setting/details**: Does the paper specify all the training and test details?
   Answer: Yes. Table 4 lists all hyperparameters; the rollout-train coordination protocol matches verl's defaults except where stated.
7. **Experiment statistical significance**: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
   Answer: Yes. All headline results report mean $\pm$ standard deviation across 5 seeds (MATH) or 3 seeds (τ-bench). Significance tests use a paired bootstrap with 10^4 resamples; we report p-values where claims of superiority are made.
8. **Experiments compute resources**: For each experiment, does the paper provide sufficient information on the computer resources?
   Answer: Yes. Table 5 reports per-experiment GPU-hours; total budget is $4{,}864$ H100-hours.
9. **Code of ethics**: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
   Answer: Yes. The work uses only publicly released model weights and benchmarks. No human subjects or personally identifiable information is involved.
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11. **Safeguards**: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?
    Answer: NA. The released artifacts are training-time corrections, not new model checkpoints with capability uplift relative to publicly available Qwen3.5.
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    Answer: Yes. Qwen3.5 weights (Apache 2.0), MATH-500 (MIT), τ-bench (Apache 2.0), verl (Apache 2.0), vLLM (Apache 2.0), FlashAttention (BSD-3) are all credited and used under their stated licenses.
13. **New assets**: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
    Answer: Yes. The GCD reference implementation and calibration script will ship with API documentation, a reproducibility README, and a JSON manifest of the calibrated constants for each Qwen3.5 scale.
14. **Crowdsourcing and research with human subjects**: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?
    Answer: NA. No human subjects research was conducted.
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    Answer: Yes. The work studies the training of LLMs themselves; all uses of LLMs are described in detail in Sections B and 6. No LLMs were used to write the paper itself in a non-standard way.